

Tower and Box: dimension analysis

Prepared: October 3, 2026. This document supplements the reconstruction report and drawings. It records source-surface variation, the initial CAD comparisons, and the final revised Tower and Box models. The last five pages include the new models, all five result figures, exact final definitions and fit comparisons. Use `Tower_revised` and `Box_revised` for the final reconstructions.

1. Establish the scale and coordinate frames

All numerical lengths below are scan coordinate units interpreted as millimetres. The known Boat reference is nominally 60 x 31 x 48 mm and gives a diagnostic scale factor 1.00043337. Tower and Box keep a scale factor of 1.0. This assumes all three files use the same export units; no independent physical unit is encoded in Tower or Box. A known physical measurement or scanner export setting is needed to establish their scale independently.

Tower's local Z follows the fitted cylinder axis, with Z=0 at the fitted lower cap. Box's local Z is normal to its outside-bottom plane, with Z=0 on that plane. X/Y define the port layout shown in the coordinate map. Rotation and translation preserve lengths. Origins and full rotation matrices are stored in `Measurements/Dimension_Measurements.json`.

```
local_point = (source_point - source_origin) @ basis
source_point = local_point @ basis.T + source_origin
```

2. Measure the original triangles

Intersect actual source triangles with horizontal planes. Tower is sampled every 0.25 mm, using 72 angular bins of 5 degrees. Each bin stores a median observed radius, with missing angles left blank. Fit section circles using equal-angle samples to reduce the influence of dense scan patches. Box wall sections avoid outlet intersections and mounting bosses. Local plane and cylinder fits reduce the influence of outliers.

The dimension CSVs separate fitted source values from CAD completions. Percentile bands describe spatial variation, including distortion and scan noise. They are not manufacturing tolerances or statistical confidence intervals. Unknown nominal dimensions, fillet radii and hidden hole depths are not recoverable merely by displaying more decimals.

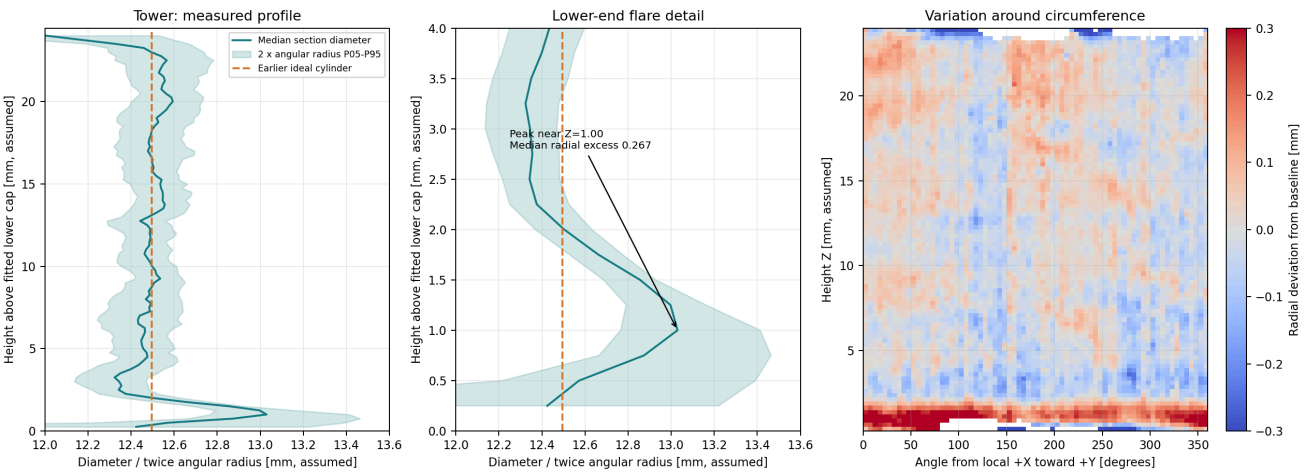
3. Distinguish envelopes from feature dimensions

Part	Original aligned vertex span X x Y x Z	Initial CAD span X x Y x Z
Tower	13.144 x 13.267 x 24.784	12.500 x 12.500 x 24.260
Box	98.556 x 90.919 x 31.836	98.125 x 90.200 x 33.151

Source envelopes include noise, missing regions and the Tower flare. Feature-based dimensions below provide the sketch dimensions needed for CAD. `Tower_reconstructed` is the initial cylinder; `Tower_measured_profile` retains the axisymmetric median flare. `Box_reconstructed` is the initial flat-floor model. Final `Tower_revised` and `Box_revised` definitions and envelopes appear in the appended results.

Tower: measured end flare and body variation

The source Tower is not a perfect cylinder. The lower end broadens around the circumference, then narrows toward the main body. The deviation map also records smaller angular and height-dependent changes.



Original-triangle sections. Shading is twice the P05-P95 angular-radius range, not a tolerance. White map cells are unobserved angles.

Quantity	Value / interpretation
Reference fitted body diameter	12.496 mm
Fitted cap separation	24.260 mm
Flare peak section Z	1.000 mm above the lower-cap datum
Median diameter at peak section	13.029 mm
Median radial projection above baseline	0.267 mm
Peak radial P05-P95	6.383 to 6.707 mm
Sampled median excess >0.10 mm band	0.750 to 1.500 mm Z
Peak angular bins above +0.10 mm	100.0% of observed bins

The band endpoints are 0.25-mm samples; they do not establish an exact fillet radius or flare start/end. Lower-cap edge coverage is incomplete, and radial asymmetry remains visible. The geometry supports an end flare, while its origin as intended design, manufacturing deformation, damage or scan distortion remains unresolved.

Tower: sketch dimensions and reconstruction choices

For the axisymmetric comparison retaining the median flare, create a sketch in a radial/height plane. Import all rows of `Measurements/Tower_revolve_profile.csv`, using radius as the horizontal sketch coordinate and Z as the vertical coordinate. Join successive points, close the profile to the axis at each end, and revolve it 360 degrees. The table below is an abbreviated view; the CSV contains every modelling knot. The final angular loft is defined in the appended revised results.

Z (mm)	Radius (mm)	Diameter (mm)	Observed angular coverage
0.000	6.248	12.496	cap completion
0.500	6.286	12.572	73.6%
0.750	6.437	12.874	84.7%
1.000	6.515	13.029	100.0%
1.250	6.499	12.997	100.0%
1.500	6.428	12.856	100.0%
1.750	6.331	12.662	100.0%
2.000	6.252	12.503	100.0%
2.500	6.171	12.342	100.0%
3.000	6.172	12.345	100.0%
4.000	6.217	12.434	100.0%
8.000	6.234	12.469	100.0%
12.000	6.243	12.486	100.0%
16.000	6.252	12.505	100.0%
20.000	6.297	12.594	100.0%
23.500	6.164	12.328	79.2%
24.260	6.248	12.496	cap completion

The Z=0 and upper-cap endpoints use the fitted body radius to complete planar caps. Some near-end sections have less than 85% angular coverage; their partial medians are explicitly labelled in the CSV. The optional revolve is axisymmetric. It retains the circumferential median flare and diameter variation, while angular deviations remain available in `Tower_angular_radius_grid.csv`.

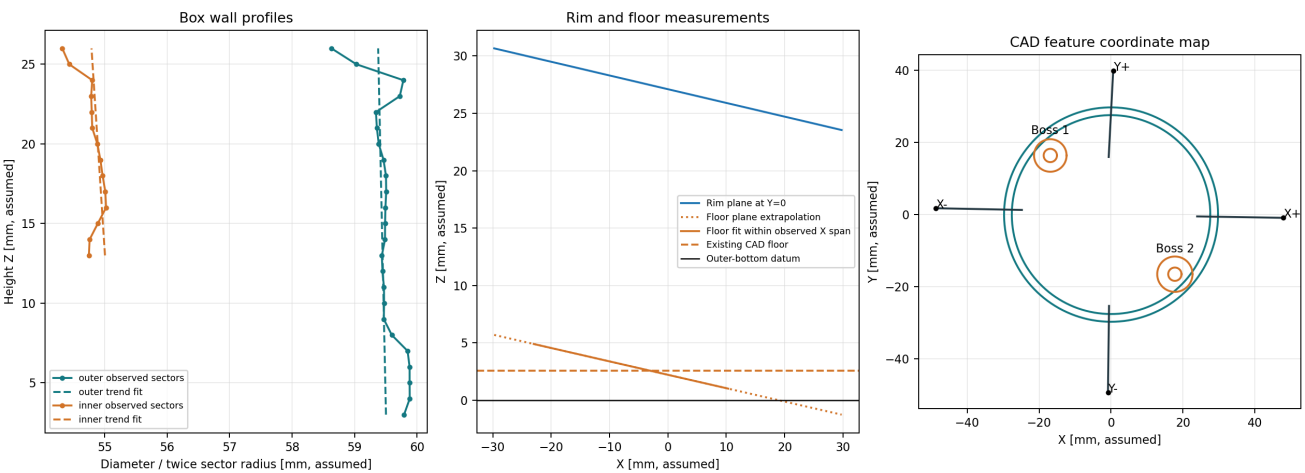
The final reconstruction uses section points $(r \cdot \cos(\text{angle}), r \cdot \sin(\text{angle}), z)$ and lofts through their angular shapes. `Tower_revised_sections.csv` gives the completed points' radii and labels. Missing bins use periodic linear interpolation; cap circles use the baseline radius. `Tower_sections.csv` also includes fitted circle centers and minimum/maximum observed opposed widths; those widths are approximate section chords, not a certified roundness measurement.

Same 18,000 source points	RMS, all points (mm)	RMS, lower-end side (mm)
Earlier ideal cylinder	0.0818	0.2088
Measured-profile alternative	0.0637	0.1126

The profile STL passes closed-topology, winding, nonmanifold-edge, degenerate-face and STL reimport checks. Agreement with observed points does not validate cap completion or establish the original design intent. Editable and exported alternatives are `CAD/Tower_measured_profile.scad`, `.stl`, and `.ply`.

Box: body, wall, floor and rim dimensions

The table below records the initial circular CAD dimensions and measured source features. Measured wall sections and fitted trends show departures from constant diameters. The final revised model uses the fitted taper trends described below. At low/high heights, scan coverage and intersections limit the available wall sectors.



Measured wall profiles and CAD feature locations. The floor fit is solid only over the observed X range near Y=0; dotted parts are extrapolation.

Feature	mm	Evidence / use
body outside diameter	59.520	fitted circular idealization
chamber inside diameter	55.220	fitted circular idealization
radial wall thickness	2.150	difference of CAD radii
cad bottom thickness	2.590	median floor estimate used by existing CAD
new floor consensus center height	2.218	independent visible-surface plane fit; different from flat CAD floor
rim center height	27.100	fitted slanted rim
rim min height on cad body	23.543	calculated from rim plane
rim max height on cad body	30.657	calculated from rim plane

The apparent outer/inner wall trends are $R_{outer}(Z)=29.759358-0.002714\cdot Z$ and $R_{inner}(Z)=27.609980-0.008295\cdot Z$. Their apparent draft angles are -0.156 and -0.475 degrees. The source residuals are larger than much of this taper, so intentional draft is not established. Initial CAD used constant radii; revised CAD uses these trends. Full source-section values are in `Box_wall_sections.csv`.

Box: sloped surfaces and coordinate definitions

The measured rim plane retained by CAD is:

```
0.118689375*X + 0.000074075*Y + 0.992931431*Z = 26.908441782
Z_rim(X,Y) = (offset - nx*X - ny*Y) / nz
```

Its slope is 6.816 degrees relative to the outside-bottom datum. At X=Y=0, Z is 27.100 mm. Use this plane to trim the body and bosses rather than assigning one height to every rim point.

An independent consensus fit to the visible inner floor gives:

```
0.115915477*X + -0.002457038*Y + 0.993256042*Z = 2.203233661
```

Its central intercept is 2.218 mm and its apparent slope is 6.658 degrees. 57,796 of 59,952 selected faces support the plane within 0.18 mm; inlier RMS distance is 0.073 mm. The fitted patch's X/Y bounds are [-22.989910474638506, -22.98980805629908] to [11.276770172289512, 22.943944526456544] mm. A bounding rectangle does not imply complete coverage inside it.

The initial CAD used a flat floor at Z=2.59 mm, derived from the earlier median estimate. This is a simplification, not a measured uniform thickness. The new plane describes a surviving patch; extrapolating it over the complete body can cross below the outside-bottom plane. The revised model resolves this using $Z_{\text{floor}} = \max(1.0, Z_{\text{plane}})$. The 1-mm minimum and socket-bore transition protecting it are explicit missing-geometry assumptions. The appended results show this completion, observed coverage and the actual exported mesh section.

All port end coordinates and boss centers on the next pages use this same Box datum. `Dimension_Measurements.json` contains the source-to-local transform so the measurements can be overlaid on the original PLY.

Box: four outlet measurements

Each port is defined by an end center E and an outward unit direction d . This retains unequal lengths, different heights and slight tilts rather than forcing four identical radial outlets.

Port	End X	End Y	End Z	Outside-body centerline length	Inclination (deg)
X+	47.920	-0.903	8.995	18.187	-2.56
X-	-48.727	1.760	14.489	19.062	4.50
Y+	0.593	39.953	11.651	10.208	-0.78
Y-	-0.859	-49.355	12.388	19.607	1.16

Lengths are mm under the shared-unit assumption. The outside-body centerline length runs from the circular body's outer wall to the outlet end along its centerline. It is not the full stock length or every visible edge length at the curved intersection.

Port	Direction X	Direction Y	Direction Z	End-plane RMS (mm)
X+	0.998867	-0.016287	-0.044717	0.064
X-	-0.996724	0.019465	0.078498	0.065
Y+	0.052114	0.998548	-0.013639	0.077
Y-	-0.012326	-0.999720	0.020215	0.063

Both CAD versions use common outside diameter 23.00 mm and bore diameter 20.00 mm. Create the outside cylinder from $E-24*d$ to E . Define the bore cutter of radius 10.00 from $E-28*d$ to $E+0.1*d$. The initial version subtracts the full cutter. For revised CAD, clip its portion inside the body to space above the completed floor, then subtract; the outside portion retains its circular bore. The 24/28-mm lengths and hidden transition are Boolean overlap/completion choices; the body intersection defines the exposed socket length.

The partial-source cylinder fits below are diagnostics. Some hit imposed fit bounds or have poor/absent inlier support, so they must not be treated as independently recovered nominal diameters. The common diameters are inferred from the better surviving outlet surfaces.

Port	Raw outside fit D	Raw bore fit D	Outer inliers	Bore inliers
X+	22.149	19.387	4434	2050
X-	26.000	20.802	472	5194
Y+	23.381	21.600	2385	1894
Y-	23.010	20.753	28976	0

X- outside fit reaches its upper radius bound; Y+ bore reaches its upper radius bound; Y- has zero bore inliers.

`Box_ports.csv` records both these diagnostics and the explicit dimensions used by CAD. Port fillets, draft and hidden bore transitions remain unresolved.

Box: mounting bosses and manual CAD build order

Boss	Center X	Center Y	Outside D	Opening D	Top-center Z	Initial center height
1	-16.90	16.41	9.04	3.70	29.119	26.529
2	17.75	-16.55	9.82	3.70	24.980	22.390

Lengths are mm. The initial boss bases start at the flat floor, $Z=2.59$. Revised boss solids join the material below the new floor, with visible height measured from the varying floor; exact revised heights appear in the appended results. Tops are clipped by the rim plane, so height varies across each boss. The first partially missing opening was completed using the second opening diameter.

Boss	Assumed hole-bottom Z	CAD centerline depth	Opening basis
1	7.00	22.119	symmetry completion from boss 2
2	7.00	17.980	fit to better-preserved upper opening

The $Z=7$ hole bottoms and corresponding depths are explicit CAD assumptions. Surviving bore fragments are discontinuous; the second boss has candidate bore faces near both the outer bottom and upper opening, but no continuous captured wall establishes a through bore. Blind/through status, countersinks and hidden depths remain unresolved. Use `Box_bosses.csv` for the definitions and observed ranges.

To reproduce the initial Box comparison model, follow this construction order:

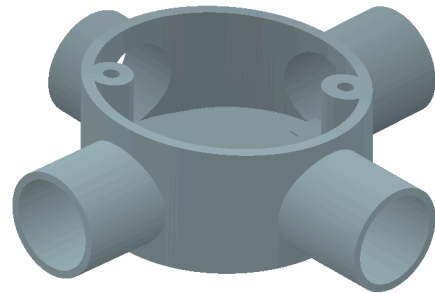
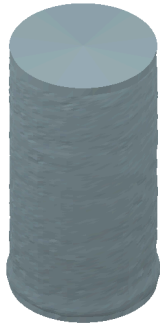
- Create the outside circular body, radius 29.76, from $Z=0$ and trim its top with the rim plane.
- Join four outlet cylinders of radius 11.50 using their measured end centers and directions.
- Subtract the central chamber, radius 27.61, beginning at $Z=2.59$, and subtract the four radius-10.00 outlet bores.
- Add the two boss cylinders at their listed X/Y centers, starting at $Z=2.59$, and trim their tops to the same rim plane.
- Cut the two radius-1.85 mounting openings from the assumed $Z=7.00$ bottoms upward.
- Check that the material boundary is one connected closed solid, while the chamber and socket bores remain open functional cavities.

The initial implementation is `CAD/Box_reconstructed.scad`. For final CAD, use `Box_revised.scad` and `revised_parameters.json`: replace constant wall radii with the measured taper functions, subtract the chamber above `max(1, Z_plane)`, protect the internal floor from bore cutters, and join bosses to the varying-floor body. Their rim and assumed $Z=7$ hole bottoms remain as above. This defines a reproducible reconstruction without establishing missing fillets, nominal dimensions, tolerances or unseen holes.

Revised Tower and Box model results

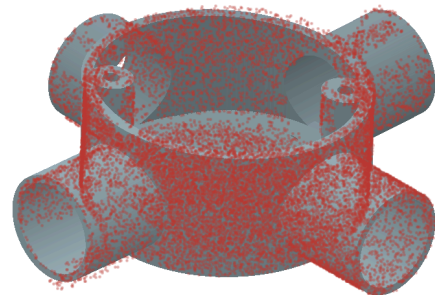
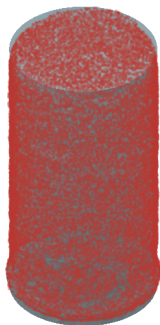
Final CAD files: CAD/Tower_revised.stl, .ply, .scad and CAD/Box_revised.stl, .ply, .scad. Boat continues to use its restored reference geometry. The earlier cylinder, median revolve and flat-floor Box are retained for comparison. All lengths assume the shared scan units are millimetres.

Final revised models and surviving source geometry
Coordinates use the same fitted local frames; source point gaps remain visible
Tower revised CAD Box revised CAD



Tower: original scan points over revised CAD

Box: original scan points over revised CAD

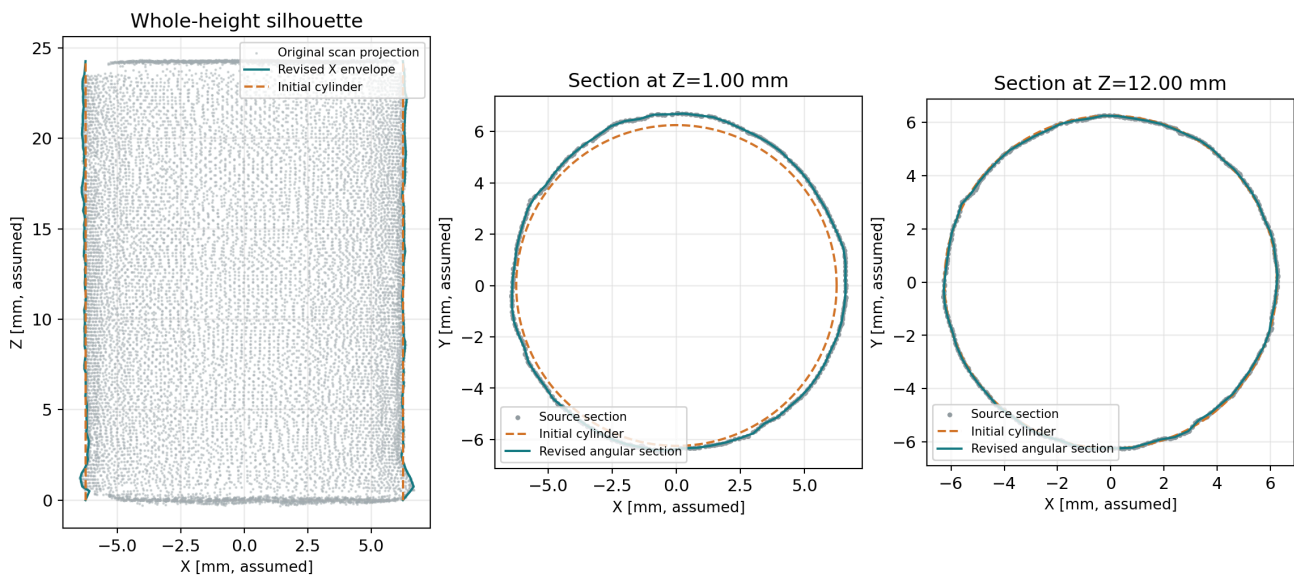


Final solids and source points in their fitted local frames. The red points are original scan samples; missing source regions are not validated by an overlay.

Part	Initial RMS (mm)	Revised RMS (mm)	RMS reduction	Revised P95 (mm)
Tower	0.0818	0.0425	48.0%	0.0825
Box	0.4772	0.3536	25.9%	0.6469

Compare exactly the same 18,000 source vertices per part against actual exported STL triangles, using seed 2026. Improvements show better agreement with captured geometry; they do not establish original design intent or recover hidden surfaces. Both revised STLs reimport as one closed material component with consistent winding, positive volume, zero open/nonmanifold edges and zero degenerate triangles.

Tower: retained flare and angular section variation



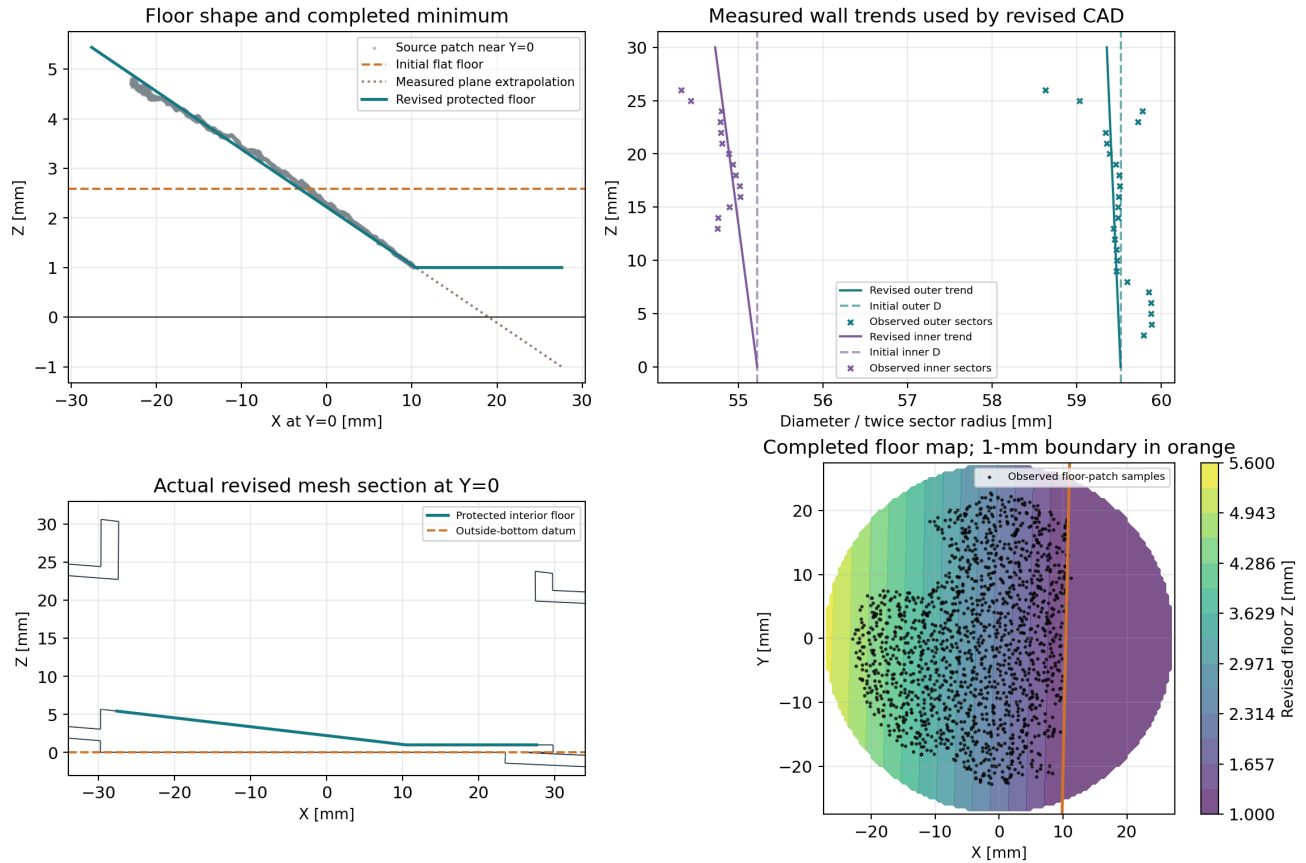
Source projection and two horizontal sections. The angular loft follows observed section shapes rather than replacing them with circles.

The final Tower connects 95 sections with 72 angles per section. Observed angular medians retain both the lower flare and circumferential variation. Unobserved bins are filled by periodic linear interpolation; the planar cap endpoints use the fitted baseline circle. Section coordinates and completion labels are in `Measurements/Tower_revised_sections.csv`; the exact radius grid is in `CAD/revised_parameters.json`.

Quantity	Revised value
Reference body diameter	12.496 mm
Median flare peak diameter	13.029 mm near Z=1
Actual X/Y/Z envelope	13.108 x 13.233 x 24.260 mm
Exported vertices / triangles	6,842 / 13,680
Geometric volume	2979.8 mm^3

The median flare diameter differs from the maximum asymmetric envelope. The source dimensions are measured; interpolation, end completion and the unknown cause of the irregularities remain reconstruction qualifications. The editable OpenSCAD source defines the loft point/radius grid and polygon faces. Face ordering follows the [OpenSCAD user manual](#). Actual STL geometry was validated; an OpenSCAD executable was not available for a separate compiler/render check.

Box: tapered walls and physically bounded floor



Measured floor patch, completed floor, wall trends and an actual mesh section. The orange map line marks the assumed minimum-thickness transition.

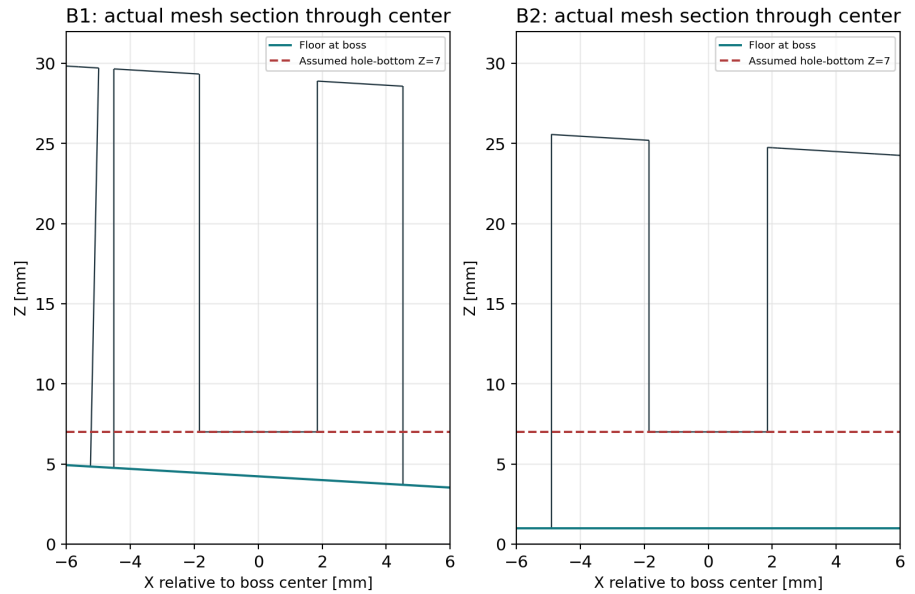
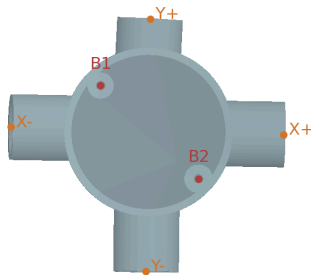
The revised body and chamber follow $R_{\text{outer}}(Z) = 29.759358 - 0.002714 \cdot Z$ and $R_{\text{inner}}(Z) = 27.609980 - 0.008295 \cdot Z$. The rim still uses the measured sloped plane. The protected floor is:

```
Z_floor(X,Y) = max(1.0, (2.203233661
    - (0.115915477)*X - (-0.002457038)*Y) / 0.993256042)
```

At the body center the floor is $Z=2.218$ mm. The 1-mm lower bound is an explicit completion assumption in missing areas, not a recovered nominal thickness. Inside the body, low socket-bore cutters are clipped to this floor; outside the body, their full circular bores are retained. The unseen transition is a conservative completion. Body-plane taper trends are also extended through unobserved wall regions; their original design intent remains unknown.

Box: outlet locations, mounting bosses and hidden completions

Box plan: outlet and boss positions



Final plan and two actual mounting-boss sections. The dashed red hole bottoms are assumed $Z=7$, not measured hidden surfaces.

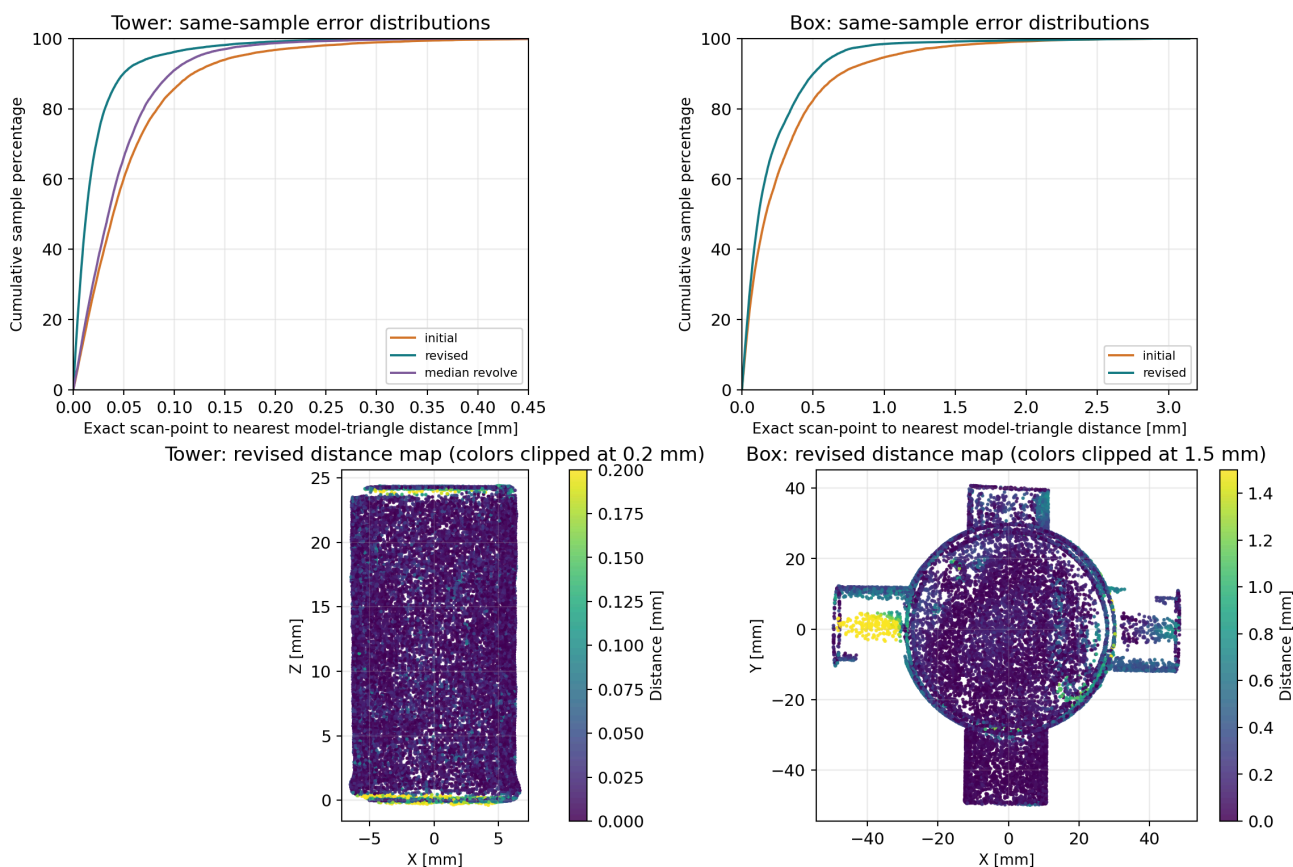
Outlet	End X (mm)	End Y (mm)	End Z (mm)
X+	47.920	-0.903	8.995
X-	-48.727	1.760	14.489
Y+	0.593	39.953	11.651
Y-	-0.859	-49.355	12.388

Outlet directions, common OD 23 / bore 20 mm, unequal end positions and overlap lengths remain the documented measured/inferred definitions. The floor-protection transition is new; hidden bore transitions and original fillet radii remain unknown.

Boss	Floor at center Z	Top at center Z	Height above floor	Assumed hole depth
1	4.231	29.119	24.888	22.119
2	1.000	24.980	23.980	17.980

Boss X/Y coordinates and outside/opening diameters remain the earlier fits. The first opening still uses symmetry and both hole bottoms remain at assumed $Z=7$. Boss solids join the material below the sloped floor; their visible height is measured from the new floor, not the old constant $Z=2.59$. The revised Box envelope is 98.125 x 90.200 x 33.141 mm. Full definitions are in `CAD/revised_parameters.json` and `Box_revised.scad`.

Validation: numerical improvement and its limits



Same-point cumulative error curves and source-point distance maps. Saturated map colors represent larger distances; table quantiles retain the full values.

Model	RMS mm	Median mm	P95 mm	P99 mm	Within 1 mm
Tower initial	0.0818	0.0394	0.1636	0.3124	100.00%
Tower revised	0.0425	0.0128	0.0825	0.1897	100.00%
Box initial	0.4772	0.1682	1.0371	1.8854	94.61%
Box revised	0.3536	0.1180	0.6469	1.4363	98.42%

Both final material boundaries pass watertightness, winding, positive-volume, single-component, degenerate-face, nonmanifold-edge and actual STL reimport checks. Vertical probes through the Box away from bosses verify the implemented sloped floor and protected 1-mm completion against the intended formula; outside-bottom hits remain Z=0.

These are one-way distances to surviving source data. Features were fitted using the same scan, so this measures reconstruction agreement, not an independent manufacturing validation. The maximum Box distance remains approximately 3.146 mm in poorly matched areas. Hidden hole depths, original fillets, missing angular regions and absolute units without the shared-export assumption remain unresolved. No physical print or dimensional inspection was performed.